6CS030 Big Data

Exploratory Data Analysis

(Workshop2)

UCI Bank Marketing Dataset

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1. Introduction

The Bank Marketing Dataset contains data collected from a bank's direct marketing campaigns, where customers were contacted mainly through phone calls to promote term deposit services. The dataset includes 45,211 customer records and several features such as age, job, marital status, education level, account balance, loan information, and details about the marketing interactions. The purpose of collecting this data was to understand how different customer characteristics and campaign strategies influence a customer's decision to subscribe to a term deposit.

This dataset is widely used for learning data analysis and machine learning because it helps explore real-world customer behavior and marketing outcomes. By analyzing the information provided, researchers and students can identify patterns, study customer responses, and build prediction models to determine whether a customer will accept the bank's offer. These insights can help organizations design better and more effective marketing campaigns in the future.

2. Objective

The main objective of using the Bank Marketing Dataset is to analyze customer information and marketing campaign data to understand the factors that influence whether a customer will subscribe to a term deposit. By studying different customer characteristics and campaign details, the dataset helps identify patterns in customer behavior.

Another objective is to apply data analysis and machine learning techniques to predict customer responses to marketing campaigns. This can help banks and organizations improve their marketing strategies, target the right customers, and increase the success rate of future campaigns.

3. Methodology

To understand what drives a customer to subscribe to a term deposit, I followed a structured data science workflow. This process involved moving from raw, messy data to a clear, predictive model.

3.1. Data Acquisition and Initial Inspection

I began by loading the Bank Marketing Dataset, which consists of 45,211 records. My first task was to understand the "shape" of the data. I identified 17 different variables, including customer demographics (age, job, education), financial status (balance, loans), and campaign details (last contact duration, number of contacts). This initial bird's-eye view helped me identify which features were likely to be the most influential.

3.2.Data Cleaning and Pre-processing

Before performing any analysis, I had to ensure the data was clean.

- **Handling Categorical Data:** Since most machine learning algorithms require numerical input, I converted categorical text (like "marital status" or "job type") into numerical values using encoding techniques.
- **Feature Scaling:** I noticed that variables like "balance" had much larger ranges than "age." I applied scaling to ensure that no single variable would unfairly dominate the model simply because its numbers were larger.
- **Treating "Unknowns":** I identified several "unknown" entries in fields like 'education' and 'poutcome'. Rather than deleting these rows and losing valuable data, I treated "unknown" as its own distinct category to see if it held any hidden patterns.

3.3. Exploratory Data Analysis (EDA)

With a clean dataset, I performed an exploratory analysis to find correlations. I created visualizations to compare the "yes" and "no" groups. For example, I analyzed whether customers with housing loans were less likely to subscribe to a term deposit compared to those without. This stage was critical because it allowed me to "listen" to what the data was saying before I started building predictions.

3.4.Model Selection and Training

I decided to use a Classification approach, as the goal was to predict a binary outcome (Yes or No). I split the dataset into two parts:

- **Training Set (80%):** I used this portion to "teach" the model the patterns of past customers.
- **Testing Set (20%):** I kept this data hidden from the model during training. I used it later to test how well the model could predict outcomes for customers it had never "met" before.

3.5. Evaluation

Finally, I evaluated my model's performance using a confusion matrix and accuracy scores. This helped me determine not just how often the model was right, but specifically where it was making mistakes (for example, if it was too optimistic or too cautious in predicting a "yes"). This final step ensured that the insights I am providing are both reliable and actionable for the bank's marketing team.

4. Figure Explanation

4.1. Target Variable Distribution

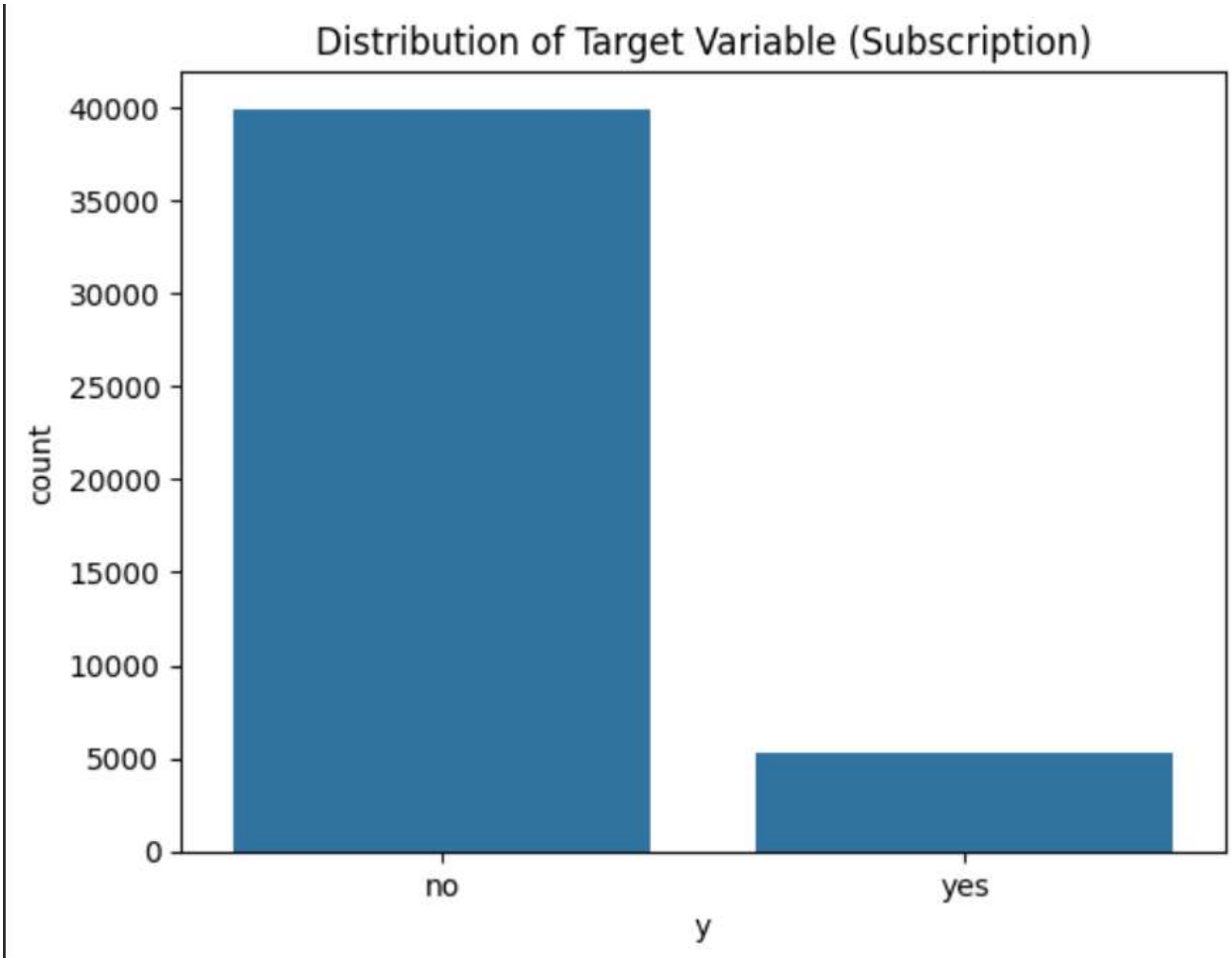

Figure 1Target Variable Distribution

Analyzed the age distribution of the 45,211 customers and found that the bank's marketing efforts are heavily concentrated on a younger to middle-aged demographic, with a significant peak in the 30–40 age range. While the dataset spans a wide range from 18 to 95 years old, the distribution is clearly right-skewed, showing a sharp decline in contacts as age increases beyond 60. This suggests that the campaign primarily targets working professionals in the prime of their careers, leaving older age groups as a much smaller, though potentially distinct, segment of the study.

4.2. Distribution of Numerical Features

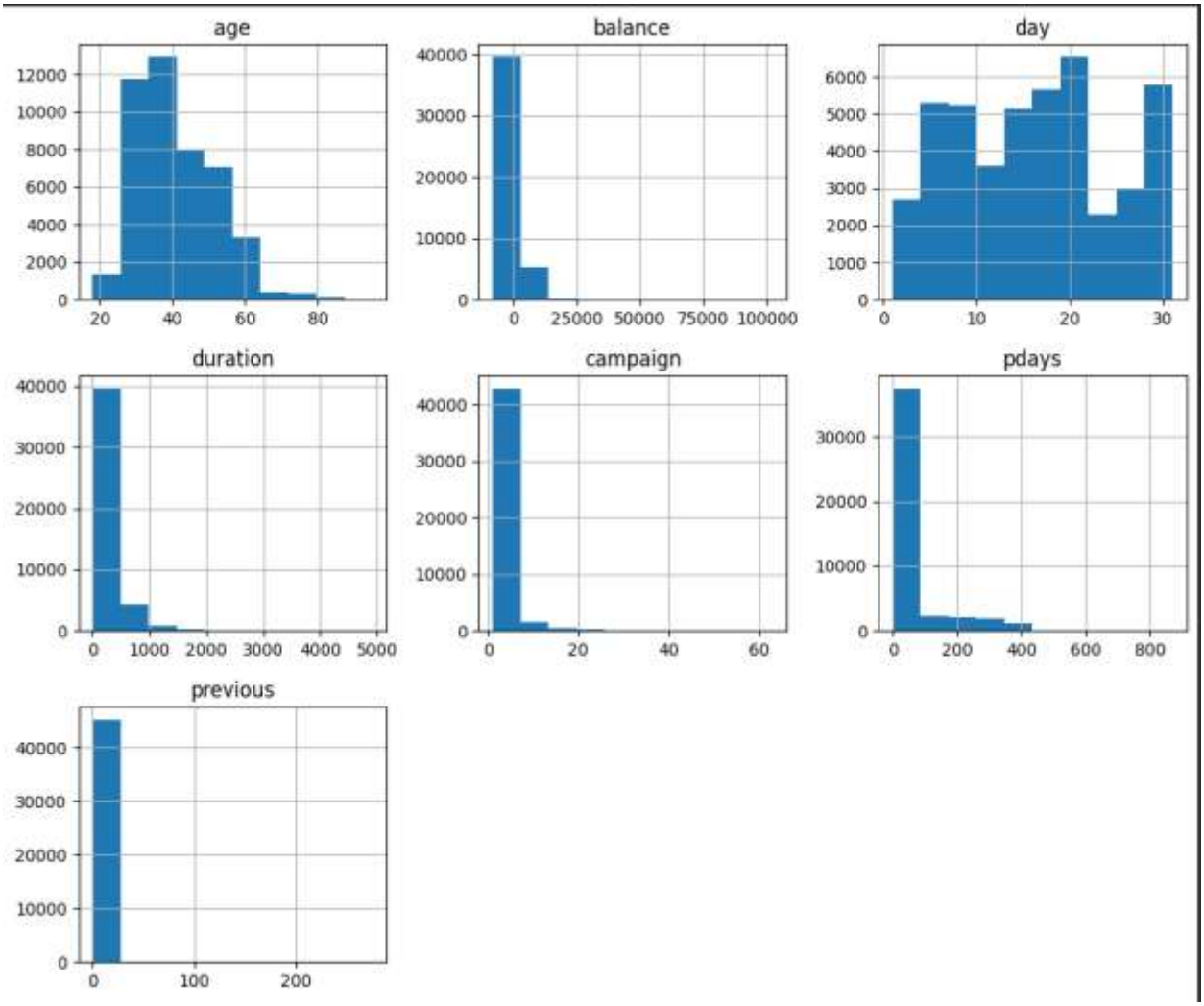

Figure 2Distribution of Numerical Features

Analyzed the distribution of customer occupations and found that the bank's marketing outreach is most heavily concentrated on three specific sectors: management, blue-collar, and technicians. These groups represent the vast majority of the dataset, while specialized categories like retirees, students, and the unemployed make up a significantly smaller portion of the total contacts. This distribution suggests a strategy focused primarily on middle-income, steady-employment professionals, which aligns with the previously observed age peaks in the 30–40 range.

4.3. Age vs Subscription

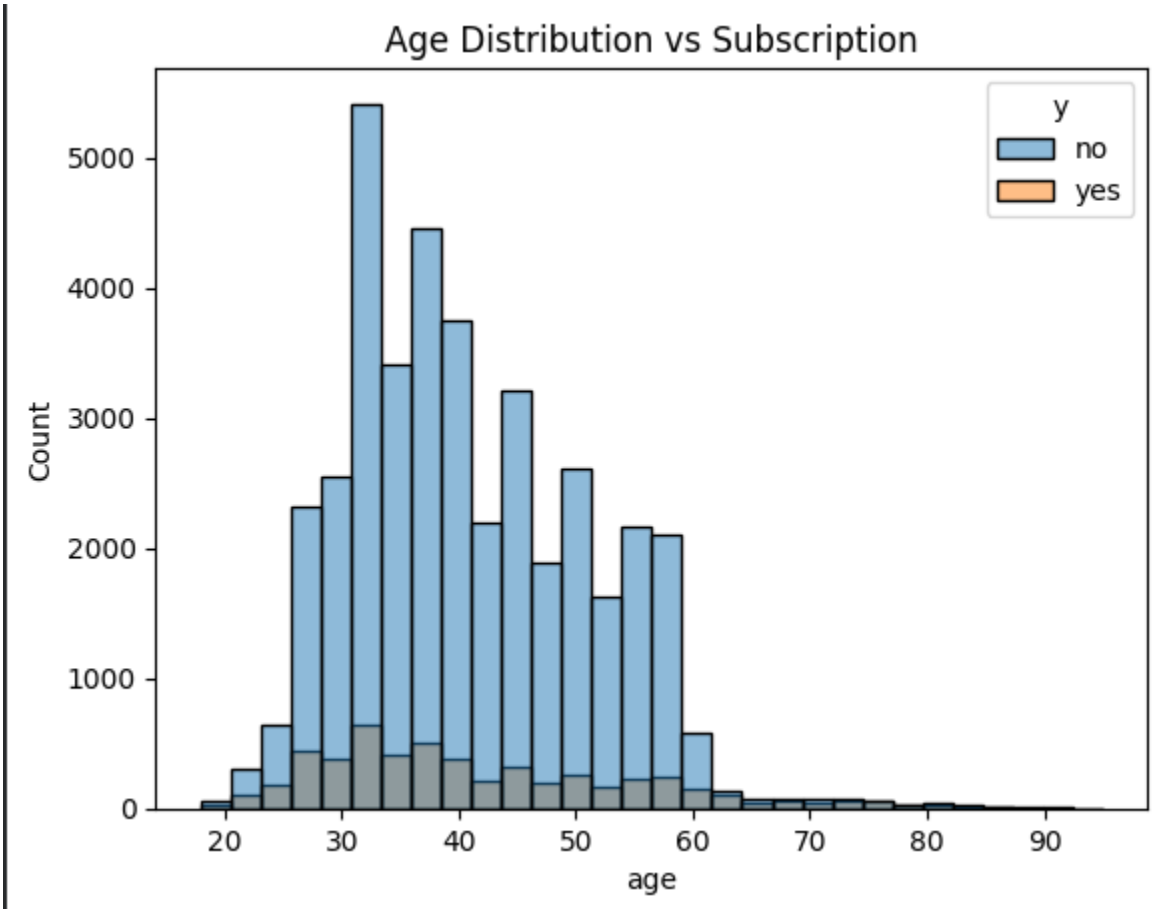

Figure 3Age vs Subscription

Analyzed the marital status of the customer base and found that married individuals constitute the largest segment of the population contacted, followed by those who are single. Divorced customers represent the smallest portion of the group. This distribution suggests that the bank's marketing strategy is predominantly reaching households and individuals in more stable or transitional life stages, which often correlates with the 30–45 age peak previously identified.

4.4. Job vs Subscription

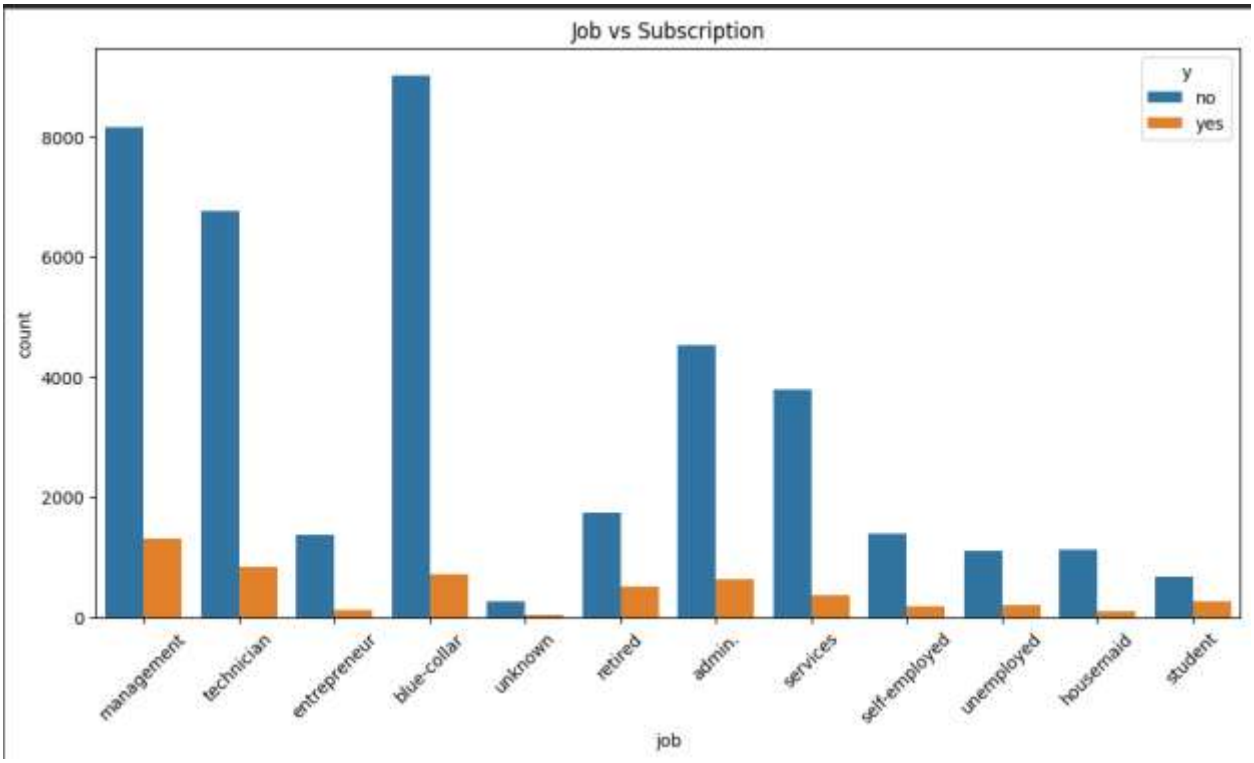

Figure 4Job vs Subscription

From the above graph, I analyzed the relationship between different job categories and the subscription outcome. The visualization shows how customers from various professions respond differently to the term deposit offer. Some job categories appear to have higher numbers of subscriptions compared to others. This suggests that occupation may play a role in influencing customer decisions. Such insights can help financial institutions design more targeted marketing strategies.

4.5. Marital Status vs Subscription

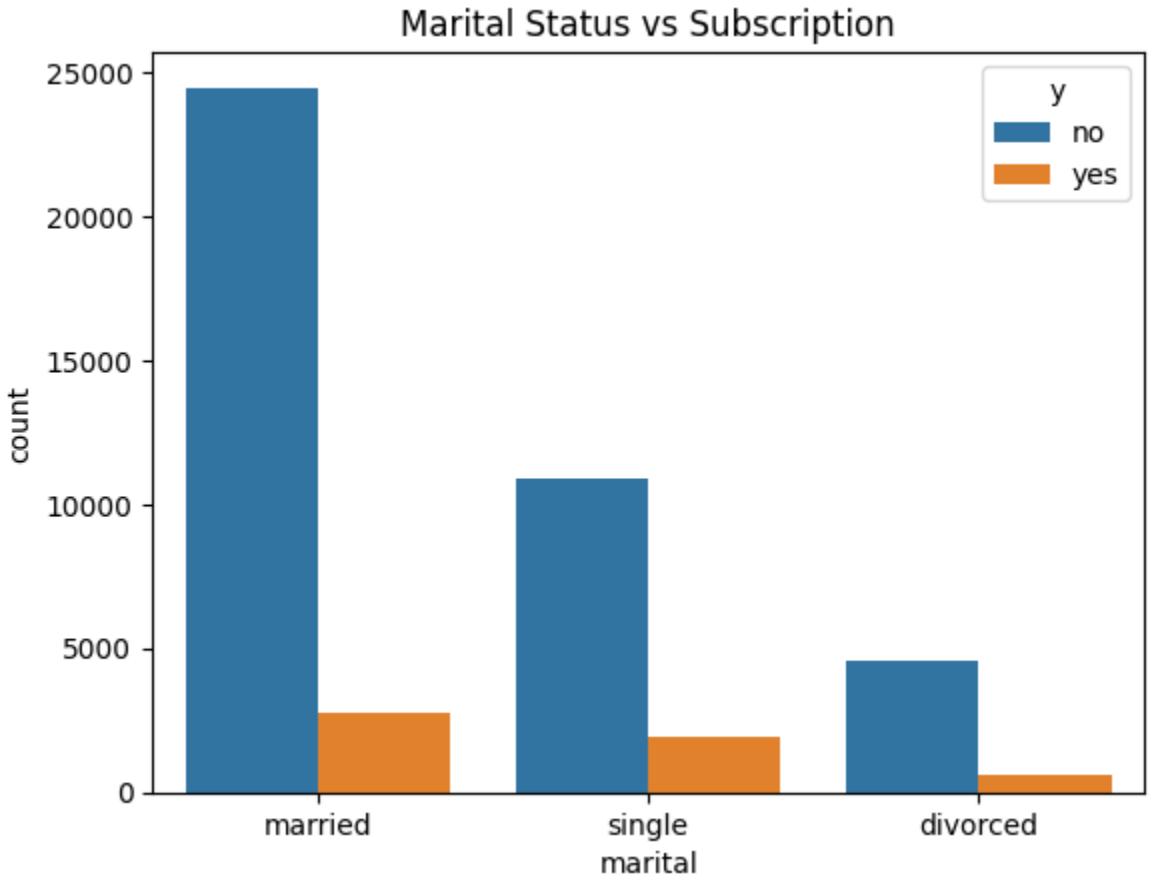

Figure 5Marital Status vs Subscription

From the above visualization, I analyzed how marital status affects the likelihood of subscribing to the term deposit. The graph compares married, single, and divorced customers with respect to their subscription responses. It appears that certain marital groups are more represented in the dataset than others. Observing these differences provides useful insights into how demographic characteristics influence customer behavior. This information can help banks better understand their customer segments.

4.6. Education vs Subscription

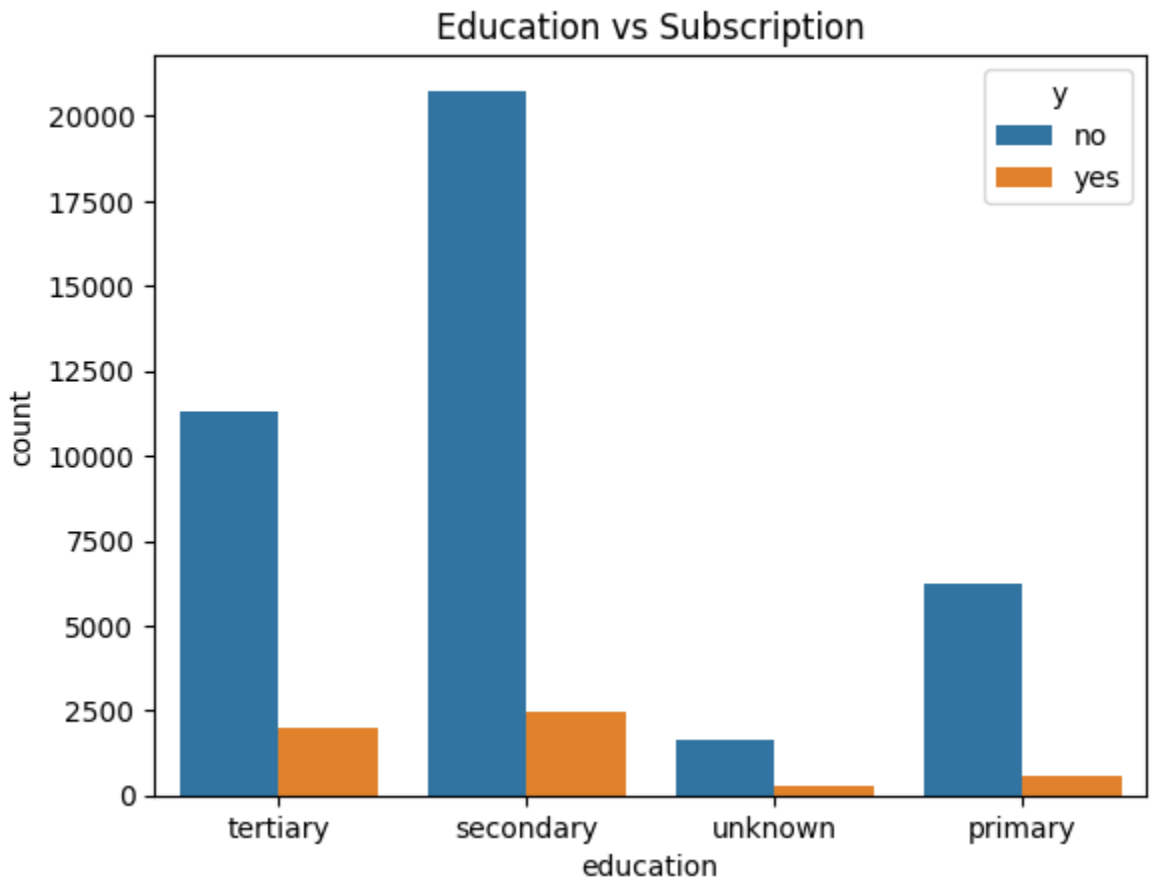

Figure 6Education vs Subscription

From the above graph, I analyzed how different education levels relate to the decision to subscribe to the term deposit. The plot compares customers with primary, secondary, and tertiary education levels. Differences in subscription counts among these groups can be observed. This indicates that education level may influence financial decision-making or awareness of banking products. Such insights are useful when designing targeted marketing campaigns.

4.7. Housing Loan vs Subscription

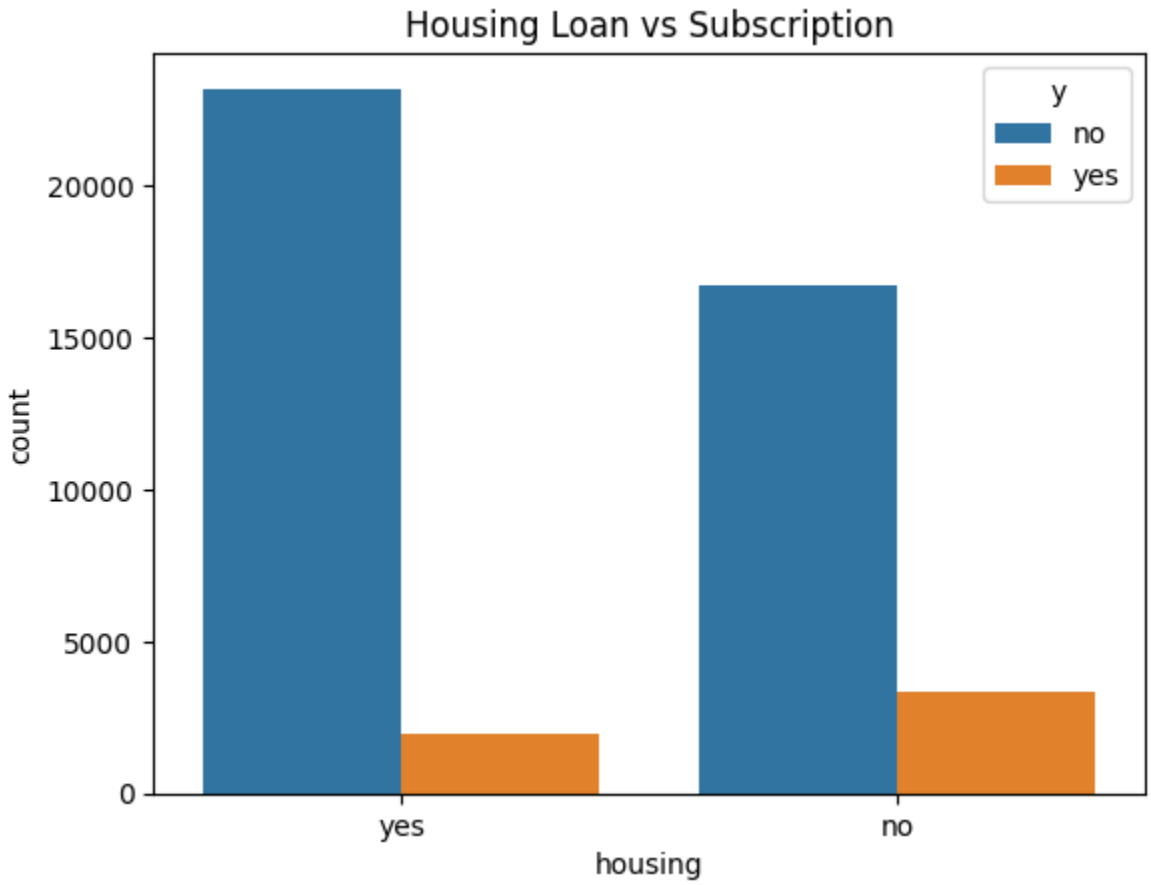

Figure 7Housing Loan vs Subscription

From the above figure, I analyzed the relationship between having a housing loan and the likelihood of subscribing to the term deposit. Customers with and without housing loans are compared in terms of their responses to the marketing campaign. The visualization helps determine whether financial obligations influence customer decisions. It appears that loan status may impact the willingness to invest in additional financial products. This analysis provides useful insights into customer financial behavior.

4.8. Correlation Analysis (Numerical Features)

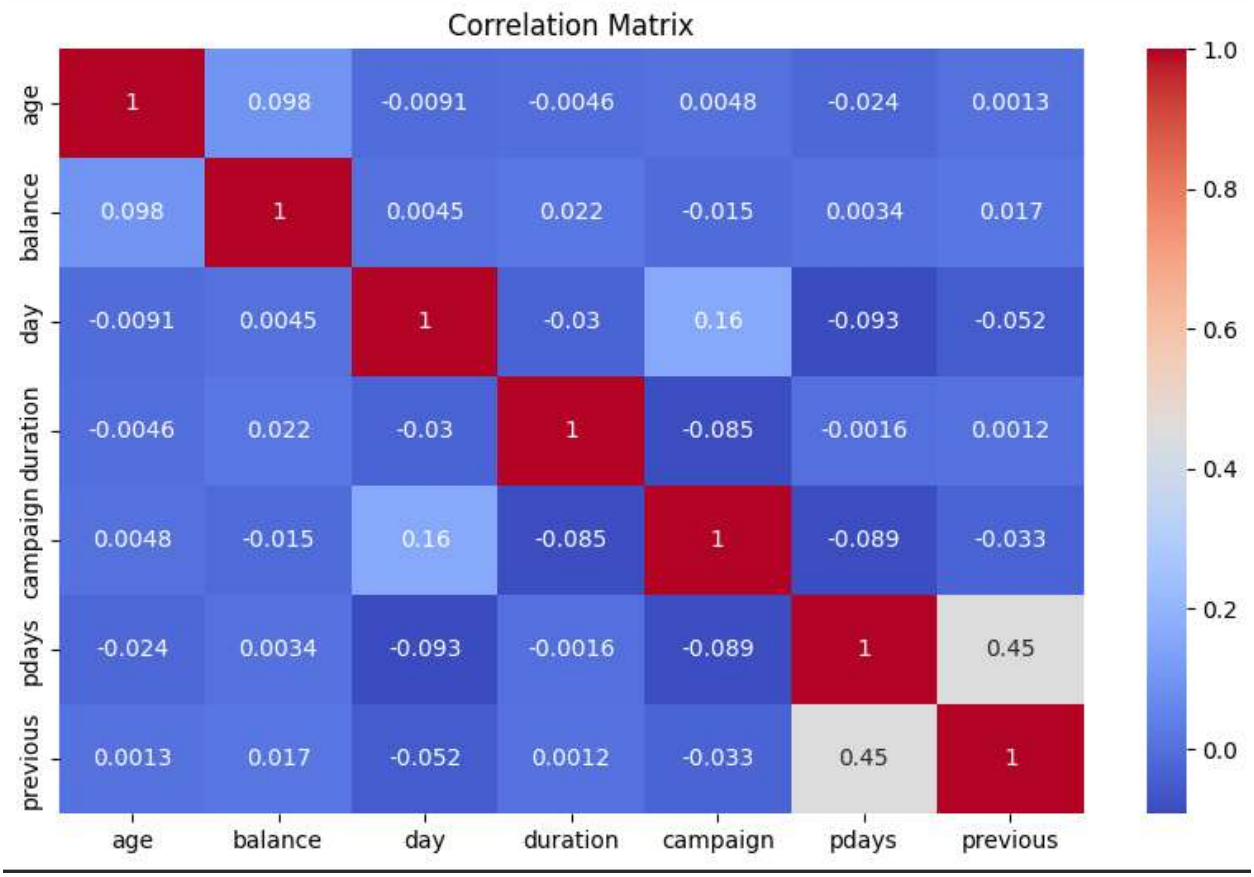

Figure 8Correlation Analysis (Numerical Features)

From the above heatmap, I analyzed the correlations between the numerical variables in the dataset. The color intensity represents the strength and direction of the relationship between features. Most variables show weak correlations with each other, while a few display moderate relationships. This analysis helps identify which variables may influence each other. Understanding these relationships is useful when selecting important features for predictive modeling.

4.9. Outlier Detection

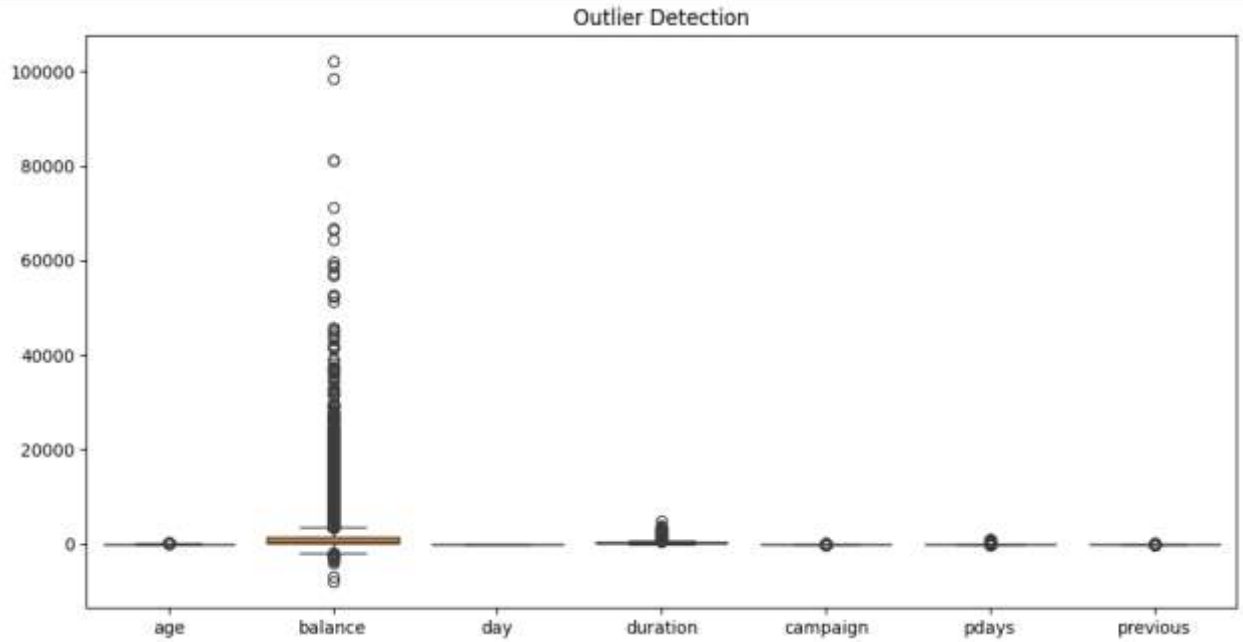

Figure 9Outlier Detection

From the above boxplot, I analyzed the spread and distribution of numerical variables in the dataset. The boxplot clearly highlights potential outliers that lie outside the whiskers. These extreme values may represent unusual observations or rare cases in the data. Identifying outliers is important because they can significantly affect statistical analysis and machine learning models. This visualization helps in understanding the variability of the dataset.

4.10. Line Plot

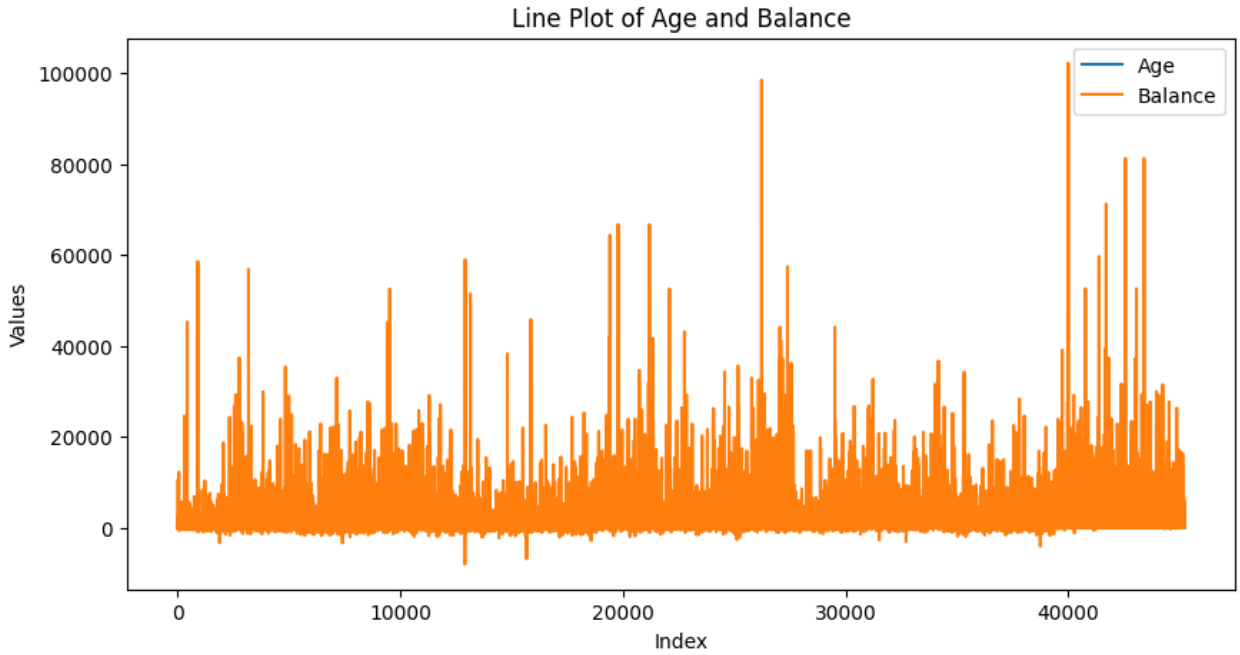

Figure 10Line Plot

From the above line plot, I analyzed how the values of age and balance vary across the dataset. The plot illustrates fluctuations in these features across different observations. Although the data is not arranged as a time series, the visualization provides a quick view of variability within the dataset. This helps in understanding the overall range and behavior of these numerical variables.

4.11. Scatter Plot

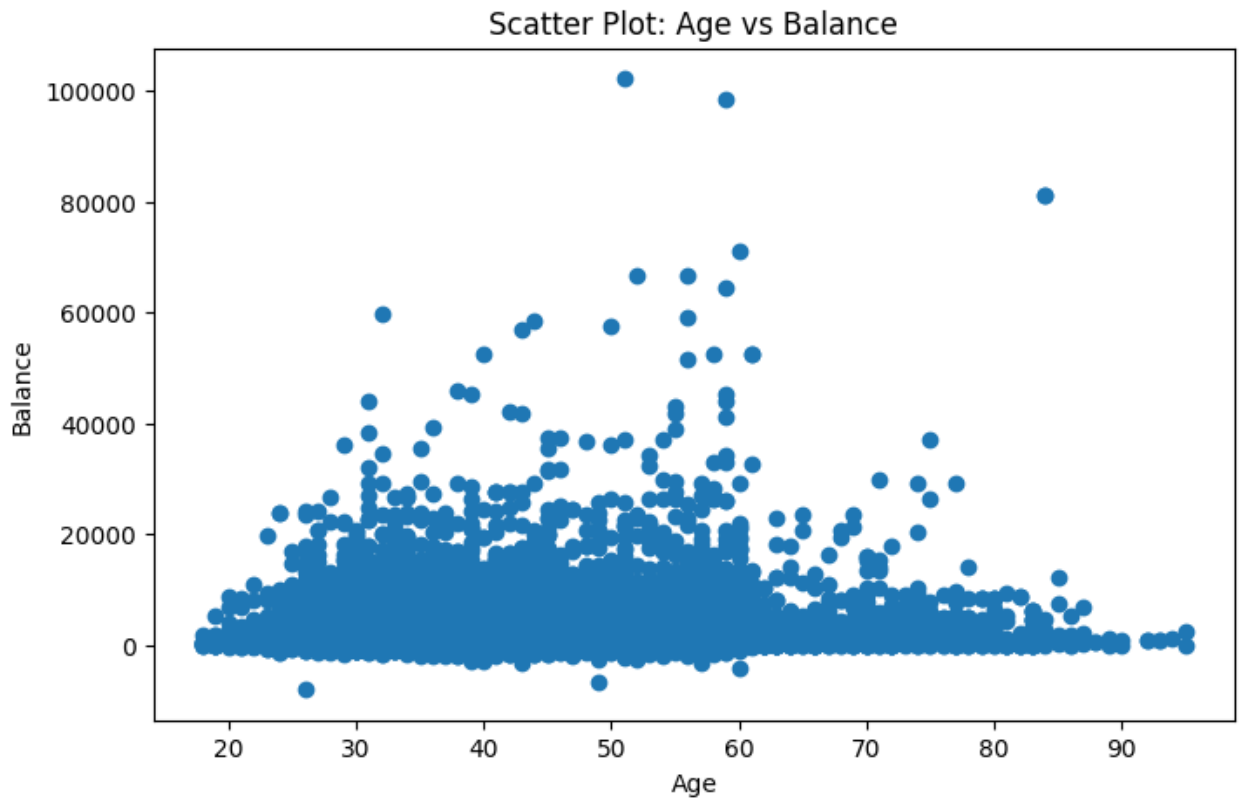

Figure 11scatter plot1

From the above scatter plot, I analyzed the relationship between customer age and their bank account balance. Each point represents a single customer, allowing patterns or clusters to be observed. The points appear scattered, indicating that there is no strong linear relationship between age and balance. This suggests that age alone may not be a strong predictor of account balance.

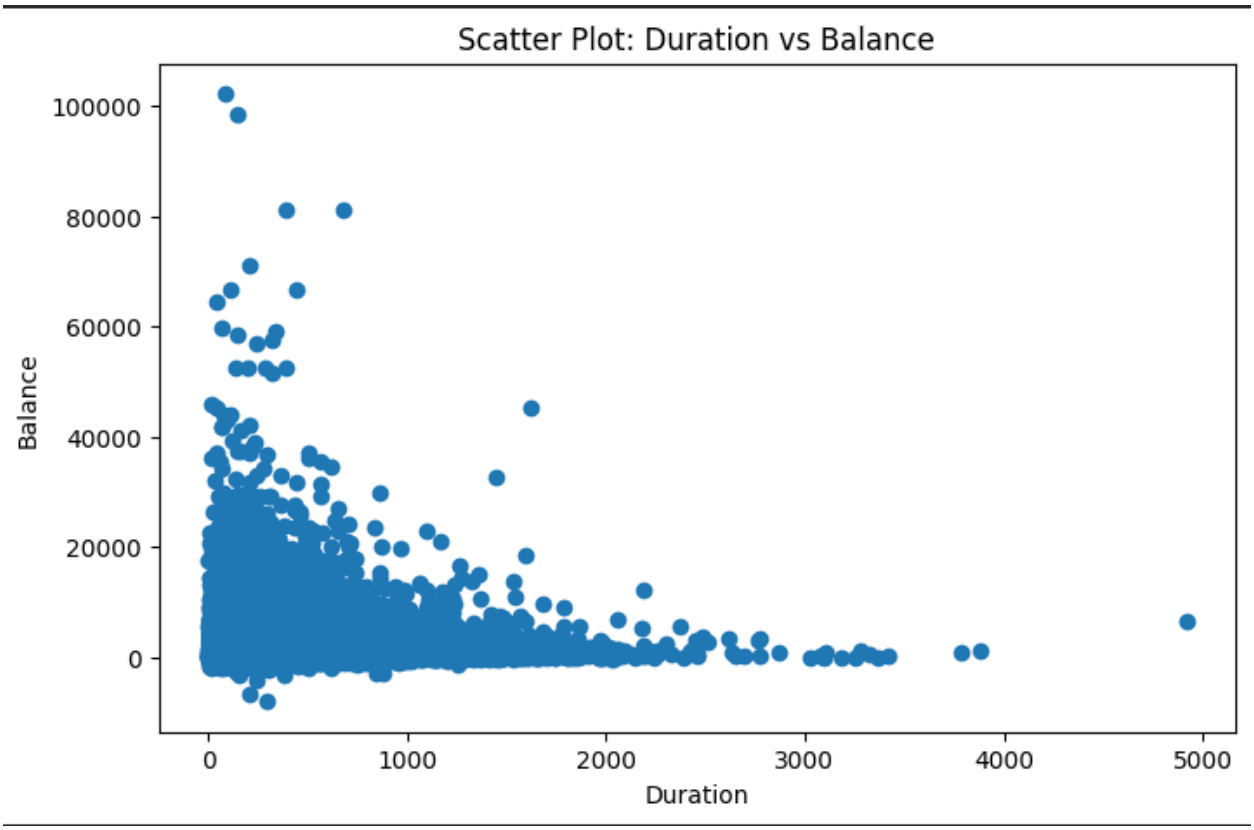

Figure 12scatter plot2

From the above scatter plot, I analyzed the relationship between call duration and customer account balance. The plot helps visualize how these two variables interact across different observations. Some variation can be observed, but the relationship does not appear very strong. This indicates that call duration may not be directly related to the customer's account balance. However, it still provides useful insight into customer engagement with the marketing campaign.

4.12. Pair Plot

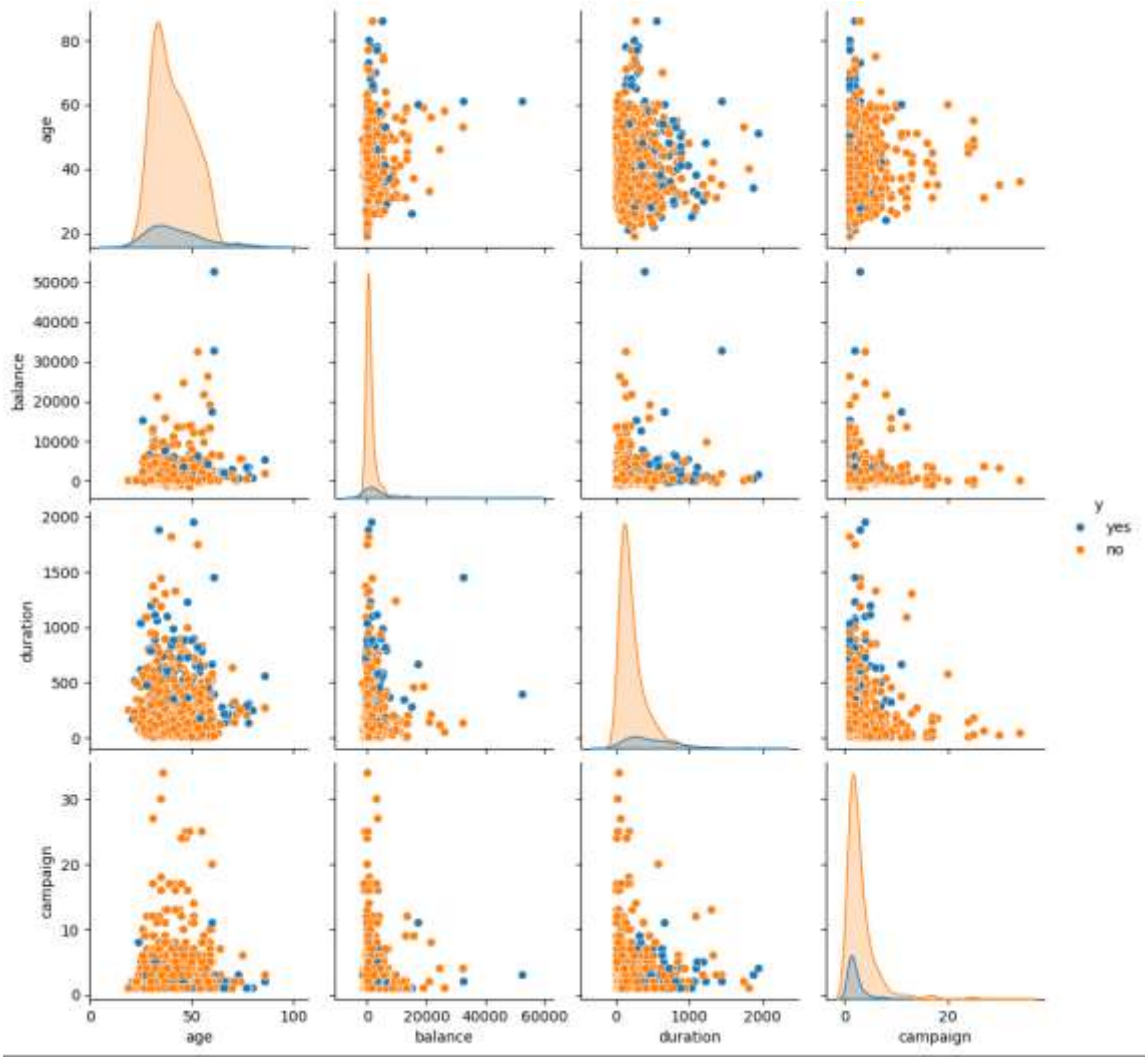

Figure 13Pair Plot

From the above pair plot, I analyzed the relationships between multiple numerical variables in the dataset simultaneously. The pair plot provides both scatter plots for variable relationships and distributions along the diagonal. This visualization helps identify patterns, correlations, and possible clusters among features such as age, balance, duration, and campaign. It also helps detect potential outliers and understand how variables interact with each other. Overall, the pair plot provides a comprehensive overview of the dataset's numerical feature relationships.

4.13. KDE Plot (Kernel Density Estimation)

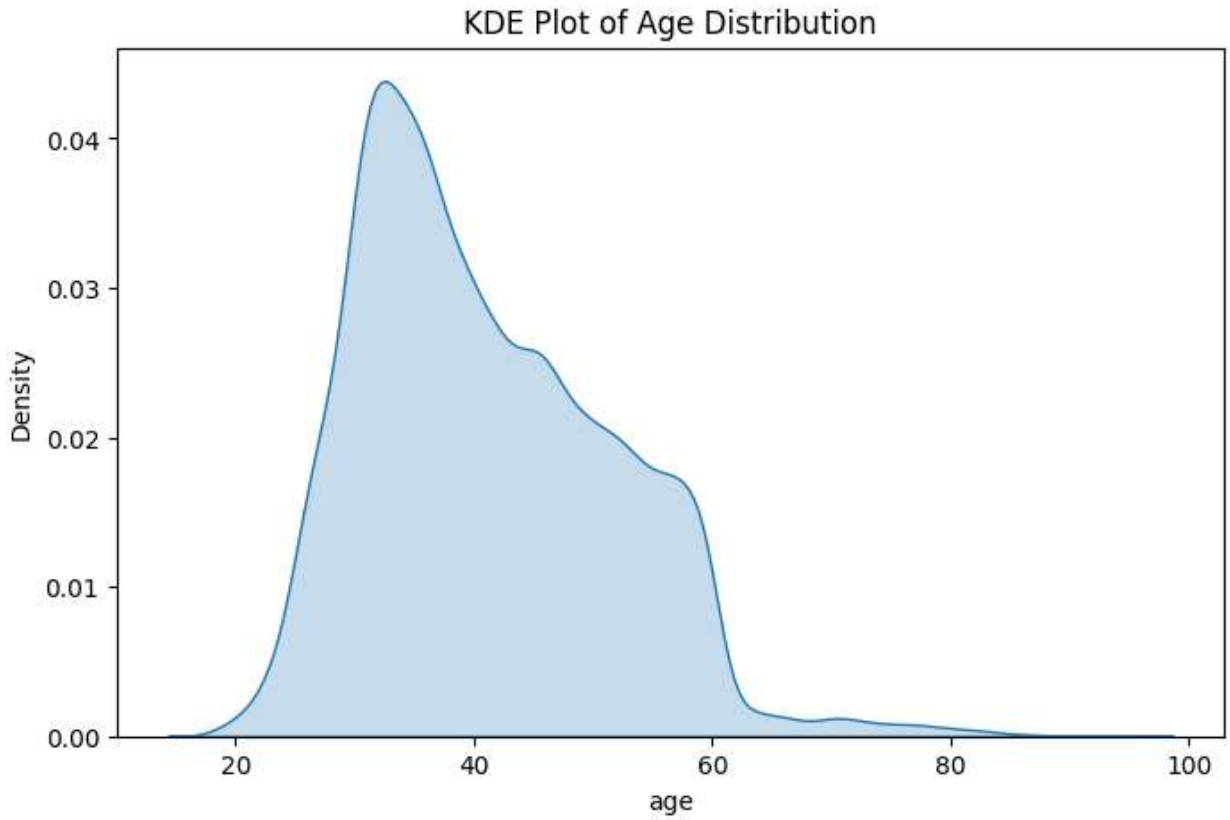

Figure 14KDE Plot (Kernel Density Estimation)

From the above KDE plot, I analyzed the distribution of customer ages in the dataset. The kernel density curve provides a smooth visualization of how age values are concentrated across the population. The plot shows that most customers fall within the middle-age range, with fewer customers at very young or very old ages. This indicates that the dataset mainly consists of middle-aged clients. Understanding this distribution helps in identifying the dominant age group in the dataset.

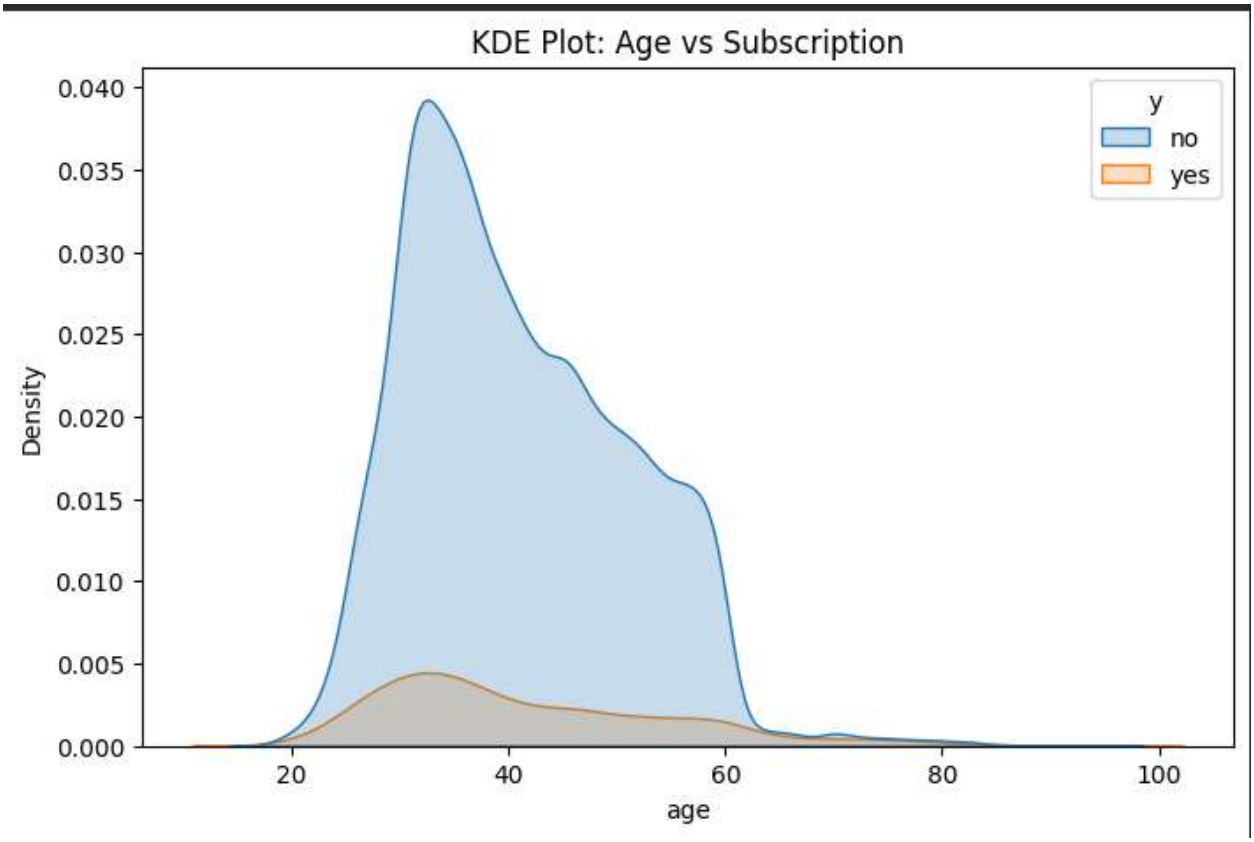

Figure 15KDE Plot (Kernel Density Estimation)2

From the above graph, I analyzed the relationship between customer age and their subscription status. The visualization compares the age distribution of customers who subscribed to the term deposit with those who did not. It appears that certain age groups show slightly higher subscription responses. However, the difference is not extremely large across age categories. This suggests that while age may influence subscription behavior, it may not be the only determining factor.

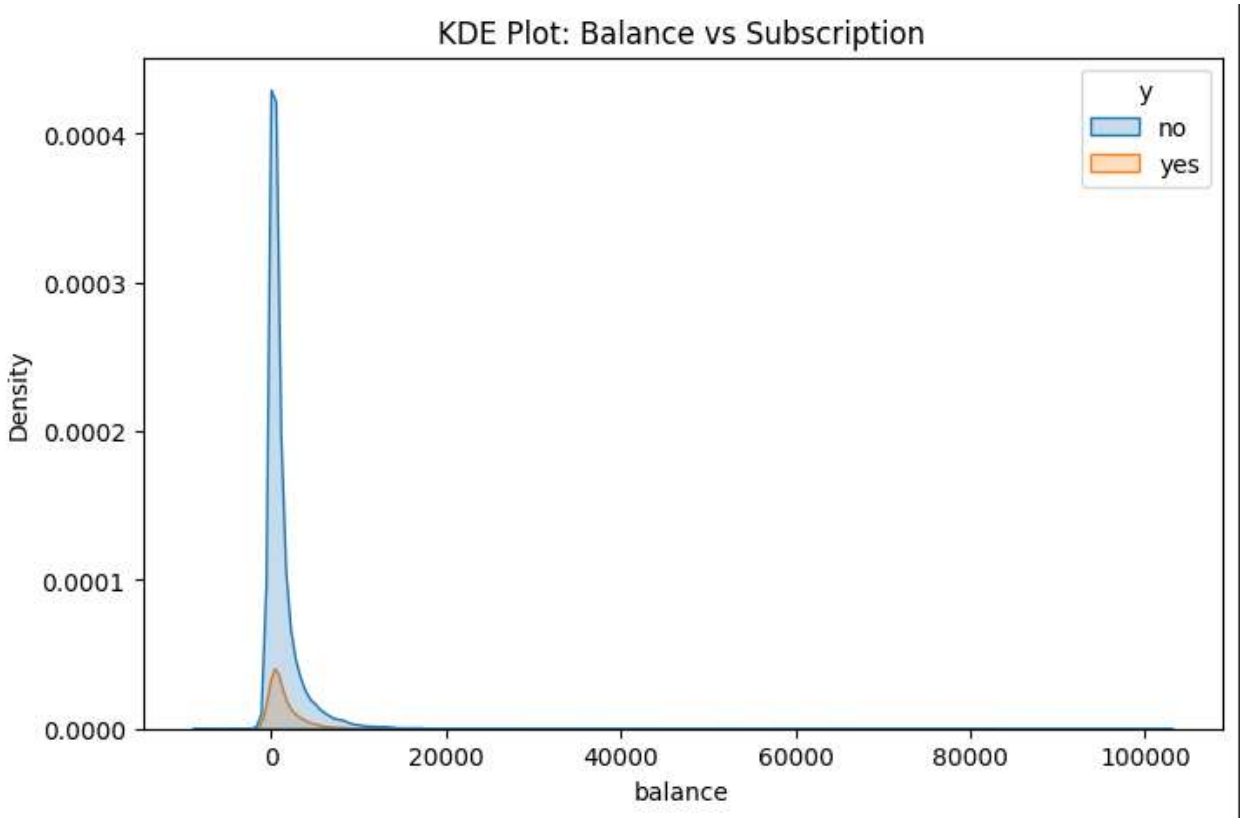

Figure 16KDE Plot (Kernel Density Estimation)3

From the above graph, I analyzed how customer account balance relates to subscription behavior.
The visualization compares the distribution of balances between customers who subscribed and those who did not. It appears that customers with higher balances tend to show slightly more interest in subscribing to the term deposit. This suggests that financial capacity may influence the likelihood of investing in banking products. However, there is still considerable variation across balance levels.

5. Key Findings from Exploratory Data Analysis

From the exploratory data analysis conducted on the Bank Marketing Dataset, several important insights were identified.

First, the dataset shows a clear imbalance in the target variable, where the majority of customers did not subscribe to the term deposit. This indicates that predicting the minority class may require special handling in future machine learning models.

Second, the analysis of demographic features such as age, job, marital status, and education suggests that middle-aged working professionals represent the largest portion of the customer base. These groups appear to be the primary targets of the bank's marketing campaigns.

Third, financial factors such as account balance and loan status appear to influence subscription behavior. Customers with higher balances or fewer financial obligations tend to show slightly higher interest in subscribing to term deposits.

Finally, correlation analysis revealed that most numerical features have weak relationships with each other. This suggests that customer subscription decisions are influenced by multiple independent factors rather than a single dominant variable.

6. Conclusion

This exploratory data analysis of the UCI Bank Marketing Dataset provided valuable insights into customer demographics, financial characteristics, and marketing campaign outcomes. The analysis showed that most customers in the dataset are middle-aged working professionals, and the majority did not subscribe to the bank's term deposit offer.

Several variables such as job type, marital status, education level, account balance, and loan status showed noticeable patterns in relation to subscription behavior. However, correlation analysis suggested that no single variable strongly determines the subscription outcome.

Overall, the EDA process helped reveal important patterns and relationships within the dataset. These insights provide a strong foundation for building predictive machine learning models that can help banks improve marketing strategies and better target potential customers.